

Technologist Academy (Engineering)

Intro

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What to expect

Ensure readiness for your first project

- 3 weeks covering software engineering fundamentals
- Local machine setup (IDE, git, node)
- Version control, design principles, API design
- Mix of theory and practical exercises
- Building backend and frontend applications and integration with databases
- Linting, caching, authentication
- Group project in weeks 2 and 3
- CI/CD
- Focus on AI-first ways-of-working

About me

- Technical architect (tech lead)
- 8 years
- Projects: HMPO, FCA, Companies House, Land Registry
- Belfast-based
- Organise AI meetups within Kainos

How can you get the most out of these sessions?

- Ask questions - if you're not sure, shout!
- Keep your own learning log
- Try and evaluate AI's outputs and understand
- Many of the practicals build upon

For group exercises:

- Communication, communication, communication...
- Work together - pair/mob programming is highly recommended
- Try and manage branches to avoid overlaps/conflicts

Week 1

Day 1:

- Ensure local dev machines are setup.
- Build your first frontend/node.js app.
- Hands-on with GitHub Copilot.

Day 2:

- Update frontend app to add new functionality (routes, services, models).
- Unit testing.
- Understand version control and how to use git.
- AI prompt engineering.

Day 3:

- Connecting frontend app to database

Day 1 agenda

(+1hr for CEST)

- 0915-1030 - intro, local dev setup, creating your first app
- 1030-1045 - **break**
- 1045-1200 - more app development
- 1200-1245 - **lunch**
- 1245-1400 - Intro to AI
- 1400-1415 - **break**
- 1415-1600 - Hands-on with AI, more app development